Reproductive Biology

Same as Previous Assessment

Arni Magnusson

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Reproductive potential

From the 2023 assessment report

The reproductive potential ogive is an important component of the assessment structure as it translates model estimates of total population biomass to the relevant management quantity, spawning potential biomass (SB).

Assumptions about maturity do not affect the process of fitting the model, only the reference point values.

The approach for calculating the reproductive potential at length ogive in this assessment is the same as the 2020 assessment.

Reproductive potential

Reproductive potential is calculated as the product of three length-based processes:

1. Proportion of females at length
2. Maturity at length (of females)
3. Fecundity at length (of mature females)

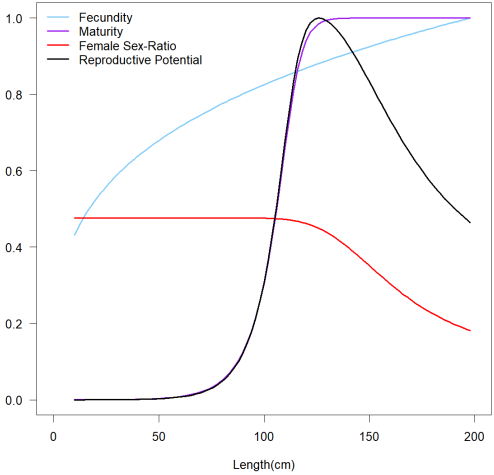
Sex ratio at length is calculated from Regional Observer Program data in SPC holdings

Maturity and fecundity are calculated as functions of length, using coefficients (Itano 2000)

$$Mat_L = \left(1 - (1-m) \times e^{-K(L-x)}\right)^{1/(1-m)}$$

$$Fecundity = a \times L^b$$

Reproductive potential



These slides:

https://github.com/PacificCommunity/ofp-sam-yft-2026-assessment